

3. Data Dictionary & Detailed Schema Specifications

3.1 BackofficeProfiles (Database: Airline_BackOfficeDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Unique backoffice staff identifier
Email	nvarchar(256)	No	UNIQUE INDEX	Staff login email address
PasswordHash	nvarchar(MAX)	No		Secure hashed password string
Name	nvarchar(150)	No		Full name of staff member
Role	nvarchar(50)	No	Default: 'Staff'	Role (SuperAdmin, Admin, HR, FinancialAdmin, Staff, Ground Staff, Dealer)
IsActive	bit	No	Default: 1	Account activation flag
ResetToken	nvarchar(MAX)	Yes		Password reset 6-digit OTP
ResetTokenExpiry	datetime2	Yes		Expiration timestamp for reset OTP
Department	nvarchar(100)	No		Department (e.g. Operations, Ticketing, Admin)
RoleTitle	nvarchar(100)	No		Job title description
AssignedAirportCode	nvarchar(100)	No		Airport base station code (e.g. DEL, BOM, BLR)
PhoneNumber	nvarchar(20)	Yes		Contact phone number
CreatedAt	datetime2	No	UTC	Record creation timestamp
UpdatedAt	datetime2	Yes	UTC	Last update timestamp

3.2 PassengerProfiles (Database: Airline_PassengerDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Passenger unique identifier
Email	nvarchar(256)	No	UNIQUE INDEX	Passenger account email
PasswordHash	nvarchar(MAX)	No		Password hash
Name	nvarchar(MAX)	No		Full display name
FirstName	nvarchar(MAX)	No		First name
LastName	nvarchar(MAX)	No		Last name
Role	nvarchar(MAX)	No	Default: 'Passenger'	User portal role
IsEmailVerified	bit	No	Default: 0	Email verification flag
VerificationToken	nvarchar(MAX)	Yes		6-digit OTP verification token
VerificationTokenExpiry	datetime2	Yes		OTP token expiry
IsActive	bit	No	Default: 1	Account status
ResetToken	nvarchar(MAX)	Yes		Password reset token
ResetTokenExpiry	datetime2	Yes		Reset token expiry
PassportNumber	nvarchar(MAX)	No		Passport document number
Nationality	nvarchar(MAX)	No		Country nationality
DateOfBirth	datetime2	Yes		Date of birth
Phone	nvarchar(MAX)	No		Contact phone number
Aadhar	nvarchar(MAX)	No		12-digit Indian Aadhar number
Gender	nvarchar(MAX)	No		Gender identity
DietaryRequirements	nvarchar(MAX)	No	Default: 'Standard'	Meal preferences (e.g. Vegetarian, Non-Veg, Jain)
MedicalNeeds	nvarchar(MAX)	No	Default: 'None'	Special medical assistance (e.g. Wheelchair)
MedicalAlerts	nvarchar(MAX)	No		Medical alerts & allergy notes

Column Name	Data Type	Nullable	Constraints / Index	Description
ProfileImage	varbinary(MAX)	Yes		Binary avatar profile picture
CreatedAt	datetime2	No	UTC	Account creation timestamp
UpdatedAt	datetime2	Yes	UTC	Profile last update timestamp

3.3 SavedPassengers (Database: Airline_PassengerDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Saved passenger primary key
PassengerProfileId	int	No	FK -> PassengerProfiles(Id)	Account owner reference
Name	nvarchar(MAX)	No		Companion passenger full name
Age	int	No		Age
Gender	nvarchar(MAX)	No	Default: 'Male'	Gender
Aadhar	nvarchar(MAX)	No		12-digit Aadhar number
PassportNumber	nvarchar(MAX)	No		Passport number
Nationality	nvarchar(MAX)	No		Nationality
DietaryRequirements	nvarchar(MAX)	No	Default: 'Standard'	Dietary preference
MedicalNeeds	nvarchar(MAX)	No	Default: 'None'	Special medical requirement
MedicalAlerts	nvarchar(MAX)	No		Medical alerts
CreatedAt	datetime2	No	UTC	Record creation timestamp
UpdatedAt	datetime2	Yes	UTC	Record update timestamp

3.4 Flights (Database: Airline_FlightOpsDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Flight template ID
FlightNumber	nvarchar(20)	No	UNIQUE INDEX	Unique flight code (e.g. AI-202)
Source	nvarchar(50)	No		Origin airport code/city (e.g. DEL)
Destination	nvarchar(50)	No		Destination airport code/city (e.g. BOM)
DepartureTime	datetime2	No		Scheduled departure datetime
ArrivalTime	datetime2	No		Scheduled arrival datetime
Gate	nvarchar(10)	Yes		Boarding gate
Aircraft	nvarchar(50)	No		Aircraft model (e.g. Boeing 737-800, Airbus A320)
Status	nvarchar(MAX)	No	Conversion: Enum	Flight status
TotalSeats	int	No		Total aircraft capacity
AvailableSeats	int	No		Remaining available seats
EconomySeats	int	No		Economy seat allocation
BusinessSeats	int	No		Business seat allocation
FirstSeats	int	No		First class seat allocation
CrewAssignment	nvarchar(MAX)	No		Assigned crew information
EconomyPrice	decimal(18,2)	No		Base economy fare
BusinessPrice	decimal(18,2)	No		Base business fare
FirstClassPrice	decimal(18,2)	No		Base first class fare
CreatedAt	datetime2	No	UTC	Record creation timestamp
UpdatedAt	datetime2	Yes	UTC	Record update timestamp

3.5 FlightSchedules (Database: Airline_FlightOpsDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Schedule instance primary key
FlightId	int	No	FK -> Flights(Id)	Reference to base flight template
DepartureTime	datetime2	No		Specific departure timestamp
ArrivalTime	datetime2	No		Specific arrival timestamp
Gate	nvarchar(10)	Yes		Gate assignment for instance
Status	nvarchar(MAX)	No	Conversion: Enum	Schedule status (Scheduled, Completed, etc.)
TotalSeats	int	No		Total schedule capacity
AvailableSeats	int	No		Available schedule capacity
EconomySeats	int	No		Remaining economy seats
BusinessSeats	int	No		Remaining business seats
FirstSeats	int	No		Remaining first class seats
EconomyPrice	decimal(18,2)	No		Schedule economy price
BusinessPrice	decimal(18,2)	No		Schedule business price
FirstClassPrice	decimal(18,2)	No		Schedule first class price
CreatedAt	datetime2	No	UTC	Record creation timestamp
UpdatedAt	datetime2	Yes	UTC	Record update timestamp

3.6 Bookings (Database: Airline_FlightOpsDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Booking primary key
UserId	int	No	INDEX	User identity (Passenger or Backoffice Dealer)
UserEmail	nvarchar(MAX)	No		Booking user email
UserName	nvarchar(MAX)	No		Booking user full name
FlightId	int	No	FK -> Flights(Id)	Referenced flight
ScheduleId	int	Yes	FK -> FlightSchedules(Id), INDEX	Referenced flight schedule
SeatClass	nvarchar(MAX)	No	Conversion: Enum	Class (Economy, Business, First)
BaggageWeight	decimal(18,2)	No		Total baggage weight in kg
PNR	nvarchar(10)	No	UNIQUE INDEX	6-character Passenger Name Record
Status	nvarchar(MAX)	No	Conversion: Enum	Booking status (Confirmed, Cancelled, etc.)
PaymentStatus	nvarchar(MAX)	No	Conversion: Enum	Payment status (Pending, Success, Refunded)
TotalPassengers	int	No		Total number of passengers in booking
ConfirmedPassengers	int	No		Number of active confirmed passengers
CancelledPassengers	int	No		Number of cancelled passengers
TotalAmount	decimal(18,2)	No		Total booking invoice amount
CreatedAt	datetime2	No	UTC	Booking creation timestamp
UpdatedAt	datetime2	Yes	UTC	Booking update timestamp

3.7 Passengers (Database: Airline_FlightOpsDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Passenger manifest primary key
BookingId	int	No	FK -> Bookings(Id), INDEX	Parent booking reference
Name	nvarchar(100)	No		Passenger full name
Age	int	No		Age
Gender	nvarchar(20)	No		Gender
AadharCardNo	nvarchar(12)	No	INDEX	12-digit Indian Aadhar
PassportNumber	nvarchar(MAX)	No		Passport document number
Nationality	nvarchar(MAX)	No		Nationality
DietaryRequirements	nvarchar(MAX)	No	Default: 'Standard'	Meal choice
MedicalNeeds	nvarchar(MAX)	No	Default: 'None'	Medical assistance requirement
MedicalAlerts	nvarchar(MAX)	No		Medical alerts
Status	nvarchar(MAX)	No	Conversion: Enum	Status (Confirmed, CheckedIn, Cancelled)
Fare	decimal(18,2)	No		Per-passenger seat fare
CancelledAt	datetime2	Yes		Cancellation timestamp if cancelled
CancellationReason	nvarchar(500)	Yes		Reason provided for passenger cancellation
SeatNumber	nvarchar(MAX)	Yes		Assigned seat number (e.g. 12A)
CreatedAt	datetime2	No	UTC	Manifest record creation timestamp
UpdatedAt	datetime2	Yes	UTC	Manifest record update timestamp

3.8 CheckIns (Database: Airline_FlightOpsDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Check-in record primary key
BookingId	int	No	FK -> Bookings(Id), INDEX	Associated booking reference
PassengerId	int	No	FK -> Passengers(Id), INDEX	Manifest passenger reference
UserId	int	No	INDEX	User identifier
FlightId	int	No	FK -> Flights(Id)	Flight reference
SeatNumber	nvarchar(10)	No		Final assigned seat number
Gate	nvarchar(10)	No		Boarding gate
BoardingPass	nvarchar(MAX)	No		Formatted boarding string (Name Flight Seat)
QRCode	nvarchar(MAX)	No		Base64-encoded QR code image data
CheckInTime	datetime2	No	UTC	Actual check-in completion timestamp
IsCheckedIn	bit	No	Default: 1	Check-in completion flag
CreatedAt	datetime2	No	UTC	Record creation timestamp
UpdatedAt	datetime2	Yes	UTC	Record update timestamp

3.9 Payments (Database: Airline_PaymentDB)

Column Name	Data Type	Nullable	Constraints / Index	Description
Id	int	No	PK, IDENTITY(1,1)	Payment transaction primary key
BookingId	int	No		Logical reference to Bookings.Id
Amount	decimal(18,2)	No		Transaction total amount
Status	nvarchar(MAX)	No	Conversion: Enum	Status (Pending, Success, Failed, Refunded)
PaymentMethod	nvarchar(MAX)	No		Method (CreditCard, DebitCard, UPI, NetBanking)
TransactionId	nvarchar(MAX)	No		Unique gateway transaction reference
CreatedAt	datetime2	No	UTC	Transaction timestamp
UpdatedAt	datetime2	Yes	UTC	Status update timestamp